

Ultra-high-frequency piecewise-linear chaos using delayed feedback loops

Seth D. Cohen, Damien Rontani, and Daniel J. Gauthier

Citation: *Chaos* **22**, 043112 (2012); doi: 10.1063/1.4766593

View online: <http://dx.doi.org/10.1063/1.4766593>

View Table of Contents: <http://chaos.aip.org/resource/1/CHAOEH/v22/i4>

Published by the [American Institute of Physics](#).

Related Articles

Faster than expected escape for a class of fully chaotic maps

Chaos **22**, 043115 (2012)

Dissipative properties of systems composed of high-loss and lossless components

J. Math. Phys. **53**, 123508 (2012)

Disorder-induced dynamics in a pair of coupled heterogeneous phase oscillator networks

Chaos **22**, 043104 (2012)

Introduction to the Focus Issue: Chemo-Hydrodynamic Patterns and Instabilities

Chaos **22**, 037101 (2012)

Predicting the outcome of roulette

Chaos **22**, 033150 (2012)

Additional information on Chaos

Journal Homepage: <http://chaos.aip.org/>

Journal Information: http://chaos.aip.org/about/about_the_journal

Top downloads: http://chaos.aip.org/features/most_downloaded

Information for Authors: <http://chaos.aip.org/authors>

ADVERTISEMENT



AIP Advances

Submit Now

**Explore AIP's new
open-access journal**

- **Article-level metrics
now available**
- **Join the conversation!
Rate & comment on articles**

Ultra-high-frequency piecewise-linear chaos using delayed feedback loops

Seth D. Cohen, Damien Rontani, and Daniel J. Gauthier

Department of Physics, Duke University, Durham, North Carolina 27708, USA

(Received 14 August 2012; accepted 17 October 2012; published online 26 November 2012)

We report on an ultra-high-frequency (>1 GHz), piecewise-linear chaotic system designed from low-cost, commercially available electronic components. The system is composed of two electronic time-delayed feedback loops: A primary analog loop with a variable gain that produces multi-mode oscillations centered around 2 GHz and a secondary loop that switches the variable gain between two different values by means of a digital-like signal. We demonstrate experimentally and numerically that such an approach allows for the simultaneous generation of analog and digital chaos, where the digital chaos can be used to partition the system's attractor, forming the foundation for a symbolic dynamics with potential applications in noise-resilient communications and radar. © 2012 American Institute of Physics. [<http://dx.doi.org/10.1063/1.4766593>]

Recently, Corron *et al.* realized a piecewise-linear (PWL) electronic circuit that simultaneously generates an analog chaotic waveform and an associated digital waveform that represents the system's symbolic dynamics. Their technique is data-efficient for chaos-based applications in that the symbolic dynamics is sufficient for specifying the underlying deterministic features of the chaotic carrier signal, but their implementation was limited to the audio frequency range (20 Hz–20 kHz). Here, we propose a new circuit architecture that operates at ultra-high-frequencies (UHF) (0.3–3 GHz). Our approach is based on two time-delayed electronic feedback loops, one analog and the other digital. Together, they ensure an increase in complexity of the analog signal and enable fast control over its dynamics, so that the overall system still behaves in a piecewise-linear fashion. Our architecture represents a potential breakthrough in the design of inexpensive and robust electric circuits that generate piecewise-linear chaos.

the systems' state variables.^{6,7} Recent studies also demonstrate that a two-state PWL operator can generate low-dimensional chaos if the system has hysteresis⁸ or uses multiple thresholds.^{9,10} Such theoretical evolutions, referred to as *PWL chaos*, are also observed in experimental systems.

Creating an experimental embodiment of a PWL chaotic system resides in the ability to design multiple linear systems (or a single system with variable parameters) and a physical switch. Examples of linear systems used to produce experimental PWL chaotic dynamics are LRC (inductance, resistance, capacitance) audio-frequency circuits.¹¹ They are popular because their parameters can be easily switched with conditions based on their voltage or current. This allows the LRC circuits to operate in two different linear states.¹² Physical switches implemented in PWL systems use various components such as digital logic gates and operational amplifiers.¹³ Though the switching times between the linear states are not infinitely fast, they are usually considered negligible for LRC circuits (being orders of magnitude faster than the system's dynamics) and thus make LRC systems good for PWL experiments with chaos.¹⁴

A novel LRC-based implementation of a PWL chaotic system has been designed recently by Corron *et al.*¹³ In their circuit, negative damping (gain greater than one) induces oscillations about a piecewise-constant voltage with a growing amplitude that is bounded by a piecewise constant driving voltage. The driving voltage switches its sign in response to a guard condition on the system's present dynamics, and leads to the generation of audio-frequency PWL chaos. Such chaos is fully characterized by the temporal evolution of the piecewise-constant voltage that serves as an easily accessible, real-time symbolic dynamics. This approach, with its simultaneous generation of analog and digital chaos, demonstrates advantages over traditional chaos generators from an application point-of-view. One of them is the existence of a matched filter for robust chaos-based communications¹⁵ and radars¹⁶ in noisy environments.

The range of applications for PWL chaos, as envisioned by Corron *et al.*, motivates an ultra-high-speed version of their circuit. However, non-ideal behaviors of LRC circuits

I. INTRODUCTION

A PWL operator with two-states is defined as

$$f(x) = \begin{cases} f_1(x), & \text{if } x < I \\ f_2(x), & \text{if } x \geq I \end{cases} \quad (1)$$

where $f_1(x)$ and $f_2(x)$ are linear operators with respect to x and I is a threshold. Every time the value of x crosses I , $f(x)$ switches between $f_1(x)$ and $f_2(x)$. This definition extends naturally to multiple states and thresholds.

Piecewise-linear operators have been used with discrete maps,¹ continuous systems,^{2,3} and to describe certain classes of electronic circuits⁴ and regulatory systems.⁵ They introduce a nonlinear component in the dynamics of systems (called PWL systems), a minimum requirement for potentially leading to chaos. For example, when a PWL operator controls the parameters of discrete maps and linear differential equations, chaos can exist in the temporal evolution of

make it difficult to realize an ultra-high-speed PWL mode-of-operation. Above a certain frequency, the propagation times of signals through LRC circuits are no longer negligible,¹⁷ and the dynamics evolves on a time scale that is comparable to the switching times of the system's electronic PWL operators. Furthermore, nonlinearities tends to arise in LRC systems at UHF^{18,19} and thus introduce undesired effects into the PWL system. We acknowledge that optical and opto-electronic systems have been shown to generate UHF chaos without such non-ideal behaviors.^{20,21} However, in order to overcome these obstacles in an all-electronic system, a new architecture must address these challenges that are at the forefront of nonlinear dynamics and electronic design.

In this paper, we present the design and operating characteristics of a novel chaotic system, shown in Fig. 1, that uses feedback loops to realize UHF chaos. In Secs. II–IV, we first describe the two main components of our electronic system: (i) an analog time-delayed feedback-loop, which generates a multi-mode oscillations; and (ii) a digital time-delayed control loop, which bounds the amplitude of the oscillations using a threshold condition and generates a digital-like waveform that follows the threshold crossings. Finally, in Secs. V–VII we present a PWL physical model and a time-delayed discrete map that capture the dynamics in our system and provide estimates for its maximal Lyapunov exponent.

II. ANALOG MULTI-MODE FEEDBACK LOOP

The time-delayed feedback loop in Fig. 1 is used to generate growing-amplitude oscillations, similar to the negative damping in designs of Corron *et al.*^{13,22} It comprises a variable gain amplifier (VGA) that is self feedback. The VGA produces

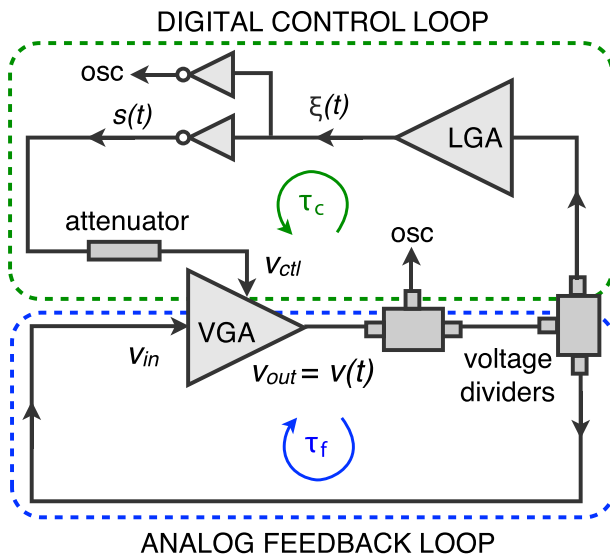


FIG. 1. Piecewise-linear chaotic waveform generator. An analog feedback loop connects the output voltage (v_{out}) of a VGA (Hittite HMC287MS8) and its input voltage (v_{in}). The VGA output also feeds a digital control loop with a LGA (Analog Devices AD8319) and a TTL (Texas Instruments SN74AUC1G04) inverter gate in series (a second TTL gate is driven in parallel in order to monitor $s(t)$ with correct impedance matching). We monitor the signals $v(t)$ and $s(t)$ using an oscilloscope (*osc*).

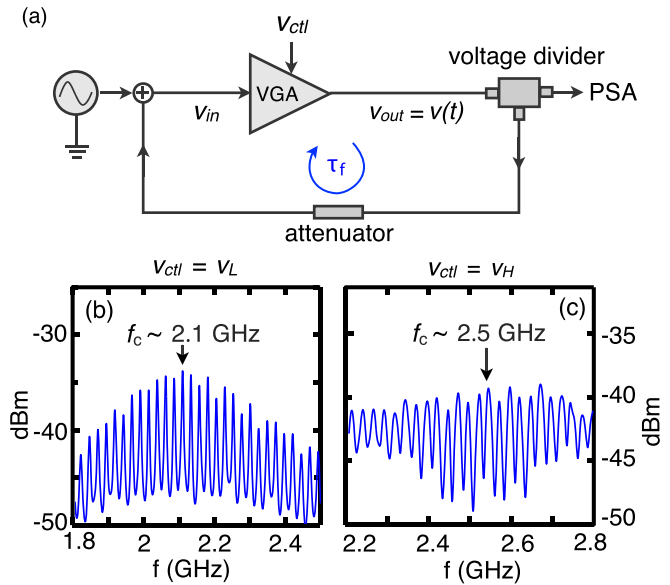


FIG. 2. Multi-mode characteristics of the feedback loop. (a) Experimental setup to determine the open-loop transfer function of the feedback loop. The input of a VGA is fed by a sinusoidal waveform generator and a delayed version of its output $v(t - \tau_f)$, where the signal $v(t)$ is monitored using a PSA. For this measurement, an attenuator of gain g_a in the feedback loop to maintain the net-gain $g_a g(t)$ below unity, avoiding self-oscillation and saturation at the resonant frequencies. In our experiment, we observe multi-mode characteristics with (b) $v_{ctl} = V_L$, the high-gain state, and (c) $v_{ctl} = V_H$, the low-gain state, where both states allow for the existence of up to ~ 20 modes (mode spacing $\sim 1/\tau_f \sim 25$ MHz).

a voltage $v_{out} = v(t)$ that is delayed by time τ_f by means of coaxial cables and connected to its input [$v_{in} = v(t - \tau_f)$]. The net gain $g(t)$ of the feedback loop depends on the VGA's gain, which is controlled by the voltage v_{ctl} applied to its control port. If $v_{ctl} > 0.5$ V then $g(t) < 1$, otherwise $g(t) > 1$. As we will show, these two gain values yield different dynamical regimes for the entire feedback loop.

The presence of a time-delayed feedback loop has important implications for the spectral properties of our system compared to that of Corron *et al.* For example, it generates multi-mode oscillations instead of mono-mode oscillations.^{13,15} Within the bandwidth B of the VGA, resonances that lead to multi-mode oscillations occur approximately at integer multiples of $1/\tau_f$. If the pass-band of B is larger than the (approximate) inter-mode spacing, the system becomes multi-mode.

In addition, the multi-mode resonances and B are dependent on the value of v_{ctl} . To characterize these spectral changes, we measure the feedback-loop transfer function for two different values of the voltage control port v_{ctl} , where the setup is shown in Fig. 2(a). It comprises a sinusoidal waveform generator with automatic frequency sweep, which drives the feedback loop, and a power spectrum analyzer (PSA) that records the output power. The experiment is first realized with $v_{ctl} = V_L < 0.5$ V, shown in Fig. 2(b). It confirms the existence of multiple modes within $B \sim 0.4$ GHz centered at $f_c \sim 2.1$ GHz. For $v_{ctl} = V_H > 0.5$ V (Fig. 2(c)), we observe resonances with relatively constant amplitude, and we approximate $B \sim 0.6$ GHz with a shift in the central frequency $f_c \sim 2.5$ GHz (see Appendix).

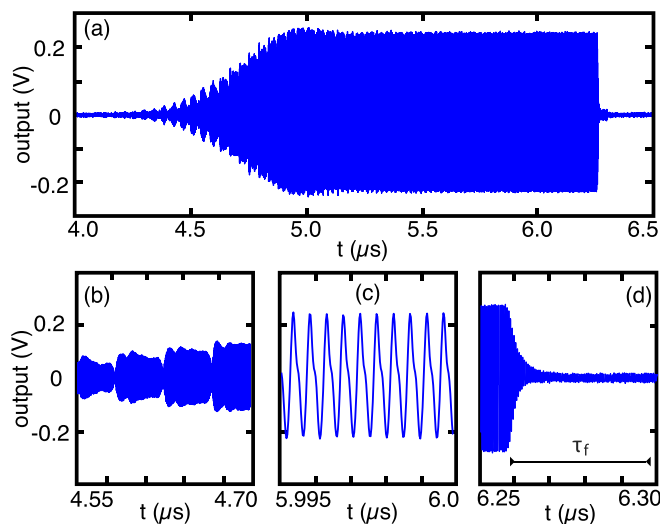


FIG. 3. Growth, saturation, and decay of $v(t)$. (a) The feedback loop is initialized with $g_H = 1.5$ at $t \sim 4 \mu\text{s}$. Electrical noise is amplified and band-pass filtered such that multi-mode oscillations emerge. Saturation occurs at $t \sim 4.85 \mu\text{s}$ and then the gain is switched low ($g_L = 0.2$) at $t \sim 6.25 \mu\text{s}$, where oscillations decay in steps of τ_f (given by a horizontal bar in (d)). Zooms of $v(t)$ showing regions where its amplitude is (b) governed by the system's multi-mode transfer function, (c) saturates, and (d) decays. Note that we choose g_H and g_L such that the rate of growth is slower than the rate of decay ($g_H - 1 < 1 - g_L$).

When the net-gain of the feedback is greater than unity (without an external source), the VGA amplifies multi-mode signals after each round-trip in the feedback loop until it saturates. Figure 3 illustrates how the feedback loop amplifies and band-pass filters electrical noise after propagating several time through the VGA with loop net-gain $g(t) = g_H > 1$. The output voltage $v(t)$ starts showing modulated oscillations (see Fig. 3(b)). When oscillations grows beyond ~ 0.3 V, the VGA saturates and behaves nonlinearly such that the dynamics becomes a stable periodic oscillation of constant amplitude at frequency $\sim f_c$ (see Fig. 3(c)). We maintain the saturated behavior for approximately $1 \mu\text{s}$ ($\sim 25 \times \tau_f$) to confirm it is not transient dynamics. Then, the net gain is switched to $g(t) = g_L < 1$, and the oscillations in the feedback loop decay until they reach the noise floor of the VGA.

This experiment shows that a single feedback-loop system can generate multi-mode UHF oscillations. Away from saturation, the feedback loop operates linearly and is governed solely by the values of its net gain ($g_{L,H}$), the time delay τ_f , and the band-pass characteristics appropriate for a given value of v_{ctl} . This experiment also shows that, without proper control over the time spent in the growth regime, the system saturates rapidly ($< 1 \mu\text{s}$), which introduces nonlinearity into the dynamics.

However, if the VGA gain is switched fast enough and with proper timing, it can avoid saturation and remain in its linear mode-of-operation. For this, in Sec. III, we introduce an additional time-delayed feedback loop that bounds the amplitude of the multi-mode oscillations. This digital *control loop* is specifically designed to parallel aspects of the guard condition from Ref. 13, where it uses a switching condition^{23,24} that produces a digital-like waveform $s(t)$.

III. DIGITAL CONTROL LOOP

We design our control scheme to vary v_{ctl} in response to a signal that monitors the amplitude of the multi-mode oscillations in $v(t)$. When the amplitude or envelope of $v(t)$ grows beyond (decays below) a set threshold, the control loop changes the value of v_{ctl} and switches the system to its decay (growth) regime. In our design, we use a logarithmic amplifier (LGA) to detect the envelope of $v(t)$ and a transistor-transistor logic (TTL) inverter gate as a digital switch (see Appendix for characterization of these components). As a whole, the control-loop generates a digital signal and is designed to prevent saturation in the feedback loop.

As shown in Fig. 1, the feedback and control loops are combined by means of a voltage divider, so that $v(t)$ drives the VGA, the LGA, and the TTL gate. The output of the TTL gate generates a digital signal $s(t)$ that is fed into a fixed voltage attenuator connected directly to v_{ctl} . We use a fixed voltage attenuator to map the low and high state from $s(t)$ ($s(t) = 0$ V, and 2 V, respectively) onto high and low values desired for $g(t)$ ($v_{\text{ctl}} = v_L$ and v_H , respectively). This design is adequate to realize a switching condition for PWL dynamics, except that it requires additional time-delay to guarantee proper control on the fast dynamics existing in the feedback loop.

With low-speed systems, the processing time of the components in a control loop, called the control-latency-time (CLT), is usually orders of magnitude faster than the time-scales of the dynamics to be controlled and therefore can be neglected.^{8,13,25} In our UHF system, however, this is not the case. The average period in $v(t)$ is 0.5 ns, which should be compared to the summed CLT of the LGA and TTL gate of approximately 9 ns. As a result, it is not possible for the switching condition to affect instantaneously the feedback-loop oscillations as they are generated.

However, we can control the envelope of the oscillations (evolving on a slower time-scale) if $s(t)$ is applied to the VGA at the proper time with respect to feedback-loop delay time. This is obtained by adjusting the time delays of both the control and feedback loops to be approximately equal. Such an approach has been already used successfully to stabilize a very-high-frequency optoelectronic chaotic system.²⁶

The time delay τ_f of the feedback loop is tunable with respect to the length of coaxial cable used in the design of our circuit. First, we guarantee that τ_f is larger than the CLT and choose $\tau_f \sim 41$ ns. Then, we use coaxial cables of time delay τ_{coax} in the control loop to approximately match the propagation time of the control signal with τ_f . We denote this net control-loop delay time by $\tau_c = \text{CLT} + \tau_{\text{coax}}$ (see Fig. 1) and adjust its value to be $\tau_c \sim 40$ ns (see Appendix for a detailed description of the total control loop CLT).

In Sec. IV, we analyze the dynamics in $v(t)$ from the full system (shown in Fig. 1) and its relation to the digital output of the control loop $s(t)$.

IV. FULL SYSTEM RESULTS

We initialize the full system with $v_{\text{ctl}} = v_L$ and $g(t) = g_H$ to ensure oscillation growth. We monitor the temporal evolution of $v(t)$ and $s(t)$ using an 8-GHz-analog-bandwidth

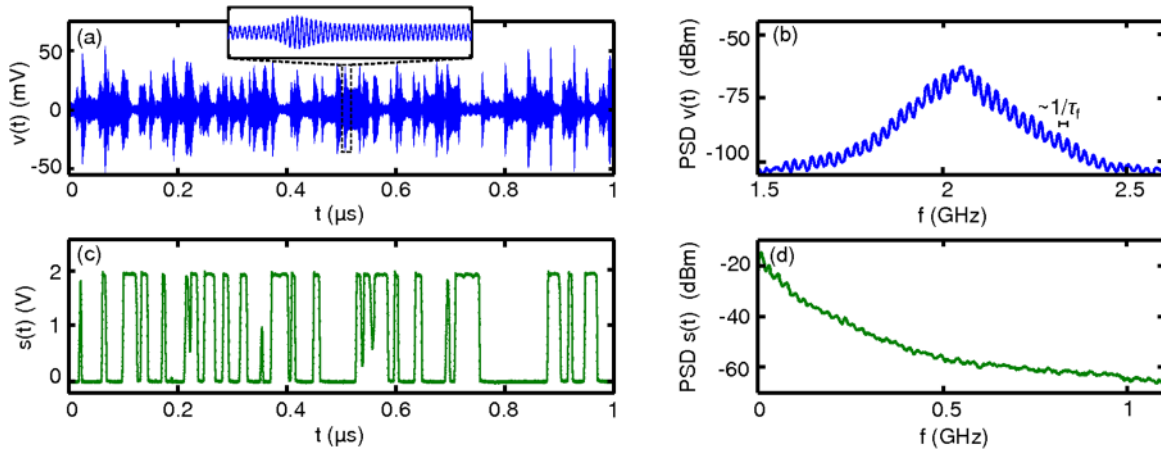


FIG. 4. Experimental time series and PSDs. (a) Time series of $v(t)$ showing UHF oscillations with complex amplitude modulations. A zoom of $v(t)$ is shown for $t = 0.5 - 0.525$ μ s. (b) PSD of $v(t)$. (c) Time series and (d) PSD of the switching state $s(t)$. The PSDs in (b) and (d) have a resolution bandwidth of 80 kHz and are smoothed over a window of 10 MHz. We note that the dynamics for this system is stable for extended periods of time, even when no thermal stabilization is used to prevent parameter drifts.

40-GS/s oscilloscope (DSO80804A) and analyze separately the time series and spectral content of both signals. Along with our analyses, we explain the observed dynamics based on the system design.

A. Experimental time series

After initial transients, the system dynamics shows oscillations in $v(t)$ with a time-varying amplitude and digital-like switchings in $s(t)$. As shown in Fig. 4(a), the evolution of $v(t)$ is aperiodic and its amplitude remains approximately between ± 50 mV, within the VGA's linear mode-of-operation. The amplitude modulation in $v(t)$ leads to spectral broadening of its power spectral density (PSD) with a peak at $\sim f_c$ and multi-mode power symmetrically distributed, as shown in Fig. 4(b). The small peaks in the PSD are spaced by $\sim 1/\tau_f$. From the peak at f_c , the PSD shows a 10-dB-bandwidth of ~ 100 MHz (consistent with the bandwidth of the LGA). Hence, the dynamics of $v(t)$ shows a UHF broadband spectrum due to aperiodic amplitude modulations.

The complexity of the dynamics in $v(t)$, which lies only in the modulation of its amplitude, has two different origins: (i) multi-mode feedback and (ii) long memory. First, noise seeds the transient oscillations of the system, which are amplified and filtered based on the system's multi-mode characteristics (see the Appendix for the initial, transient dynamics). Then, these transient amplitude modulations are stored in the system's memory as they propagate around the feedback loop and continuously affect the future dynamics at time $t + \tau_f$, causing the dynamics to reside in an infinite-dimensional phase space and allowing for the existence of chaos.²⁷

Simultaneously, the control loop generates $s(t)$, a digital-like, asynchronous (unclocked) voltage. It switches aperiodically between its low and high states with minimum pulse widths of approximately 10 ns (see Fig. 4(c)). The finite bandwidth of $s(t)$ is visible on the experimental PSD spanning frequencies from dc to ~ 100 MHz, the cutoff frequency at -10 dB (see Fig. 4(d)). The signal $s(t)$ also shows short-pulse rejection,²⁸ a mechanism by which the TTL gate generates

intermediate voltages in response to fast fluctuations at its input. As a result, our system shows slight deviations from ideal PWL behavior, discussed in Subsection IV B.

We finish this section by comparing the time series of $v(t)$ and $s(t)$. Unlike the audio-frequency PWL system presented in Ref. 13, there is no direct correlation between the analog and digital dynamics. This feature is unique to our system and is caused by the multi-mode feedback loop. In Subsection IV B, we construct two and three-dimensional (2D and 3D) projections of the system's attractor. We demonstrate a relationship existing between $v(t)$ and $s(t)$, and we also examine the probability distribution of the feedback-loop gain to highlight the non-ideal behaviors of the circuit with respect to a perfectly PWL system.

B. Projection of experimental attractor

We examine projections of the system's attractor using time-delay embeddings of the dynamics. In our system shown in Fig. 1, we choose the embedding time based on the relation

$$v(t) \sim g(t)v(t - \tau_f). \quad (2)$$

The approximate relation results from the filtering characteristics of the VGA. Similar to Ref. 13, we also discretize Eq. (2) and use the local maxima in $v(t)$, labeled as v_n , where n is an integer incremented over each maxima that are spaced by $\sim 1/f_c \sim 0.5$ ns. These local maxima contain the information about the system's amplitude modulations. Within the interval $[t, t + \tau_f]$, there are on average $T_f = 87$ local maxima, and therefore, using the discrete embedding time T_f , we can construct attractor projections.

A 2D projection of the system's attractor is represented graphically by examining the pairs (v_{n-T_f}, v_n) . Figure 5(a) shows the projection with two different clusters of points with finite widths and slopes that coincide approximately with the gains g_H and g_L (labeled accordingly in the figure). The linearity of these clusters is expected because of the relation $v_n \sim g_n v_{n-T_f}$, where $g_n = g(t_n)$ is the gain at discrete

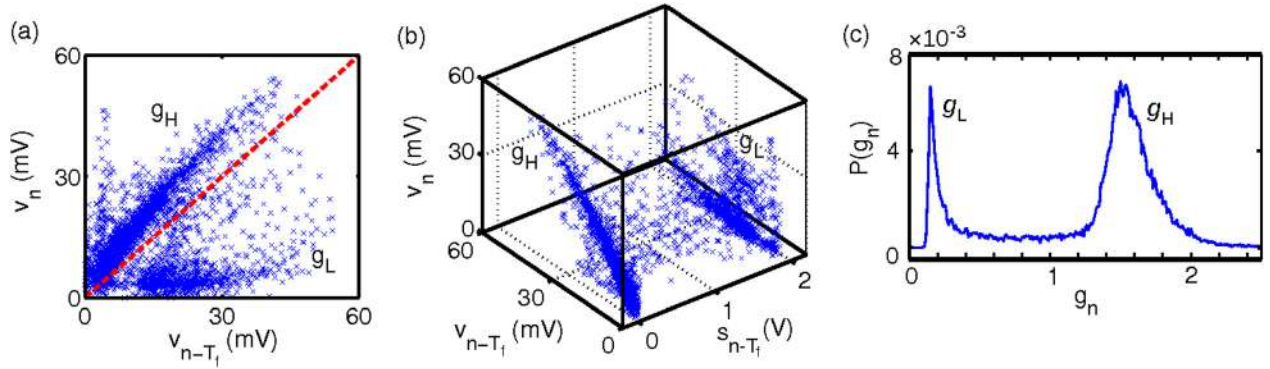


FIG. 5. Projections and analysis of the experimental attractor. (a) 2D projection using v_n and v_{n-T_f} with two approximately linear clusters of points labeled by $g_{L,H}$. A dashed line with unity slope serves as a reference. (b) 3D projection using v_n , v_{n-T_f} , and s_{n-T_f} , demonstrating that s_n partitions the two clusters. (c) Probability density $P(g_n)$ of the gain states with peaks labeled $g_{L,H}$.

times t_n . The finite widths of the clusters relate to small over and under shoots in the values of the gain (due to filtering) and intermediate $s(t)$ values applied to v_{cl} .

In addition, depending on the state of $s(t)$, the map's dynamics resides on one of its two clusters, respectively. Using $s_n = s(t_n)$, we construct a 3D projection of $(s_{n-T_f}, v_{n-T_f}, v_n)$ in Fig. 5(b). Every time s_n switches, the dynamical state (v_{n-T_f}, v_n) transitions from one cluster to the other, demonstrating that s_n partitions the system's attractor (See the Appendix for a preliminary analysis of the alphabet in the symbolic dynamics and its underlying grammar).

We also analyze the attractor by looking at the statistics of $g_n \sim v_n/v_{n-T_f}$. In Fig. 5(c), the probability density of the gain $P(g_n)$, is given as a function of g_n and shows two primary peaks that occur at approximately $g_{L,H}$, the most probable values of g_n . There is also a higher probability to have $P(g_n) > 1$ because of the larger area under the g_H peak, which indicates on average more growth than decay in the dynamics. The spread of the peaks is due to the finite bandwidth of the VGA, where v_n cannot rise or fall infinitely fast. Finite bandwidth is a limiting property of all high-speed electronics, and is the main cause of non-ideal PWL behaviors in our system. We now verify these non-deal behaviors using a continuous model of the dynamics.

V. PIECEWISE-LINEAR PHYSICAL MODEL

In this section, we model both the feedback and control loops using delay differential equations (DDEs). All of the model parameters are obtained from regression analysis using experimental data as detailed in Appendix.

A. Multi-mode feedback loop model

We model the feedback loop as a band-pass filter with time-delayed feedback and gain²⁹

$$\frac{\dot{v}(t)}{\Delta(t)} + v(t) + \frac{(\omega^{(o)}(t))^2}{\Delta(t)} \int v(t') dt' = g(t)v(t - \tau_f), \quad (3)$$

where τ_f is the feedback-loop time delay and $\Delta(t)$, $\omega^{(o)}(t)$, and $g(t)$ are *approximately piecewise-constant* (APC) parameters that continuously switch. We note that Eq. (3) is completely linear for fixed $\Delta(t)$, $\omega^{(o)}(t)$, and $g(t)$, as we have defined the

feedback loop without a nonlinear saturation term. Therefore, for $g(t) > 1$, Eq. (3) becomes unstable and $v(t)$ diverges. As we will discuss in Subsection VB, the control loop, which continuously switches the values of the APC parameters, is the only mechanism that keeps $v(t)$ from diverging (experimental saturation).

The first two APC parameters are modeled as $\omega^{(o)}(t) = 2\pi\sqrt{f^{(+)}(t)f^{(-)}(t)}$ and $\Delta(t) = 2\pi(f^{(+)}(t) - f^{(-)}(t))$, where $f^{(+)}$ and $f^{(-)}$ are the upper (+) and lower (−) cutoff frequencies of the multi-mode band-pass filter (−3 dBm drop off), respectively. In the model, these cutoff frequencies shift depending on the switching state $s(t)$ as

$$f^{(+,-)}(t) = \frac{s(t)}{A} (f_L^{(+,-)} - f_H^{(+,-)}) + f_H^{(+,-)}, \quad (4)$$

where $f_H^{(+)} = 2.41 \pm 0.01$ GHz, $f_H^{(-)} = 1.85 \pm 0.01$ GHz, $f_L^{(+)} = 3.29 \pm 0.1$ GHz, and $f_L^{(-)} = 1.97 \pm 0.04$ GHz and $s(t)/A$ is governed by the model for the control loop. Thus, for example, when $g(t) = g_{L,H}$, $\Delta(t) = \Delta_{L,H} = 2\pi(f_{L,H}^{(+)} - f_{L,H}^{(-)})$.

The last APC parameter is the gain $g(t)$, which is also dependent on $s(t)/A$ through the relation

$$g(t) = \frac{s(t)}{A} (g_L - g_H) + g_H, \quad (5)$$

where $g_H = 1.5$ ($g_L = 0.2$) is the high (low) gain state. In our model, all of the APC constants have the same rise and fall times (instead of infinitely fast switches) that follow the physical switching state $s(t)$. This approximation captures some of the non-ideal behaviors of the experiment while maintaining approximate PWL dynamics.

B. Control loop model

The control loop is modeled using two separate nonlinear operators to approximate the LGA and TTL gate. The LGA first rectifies a delayed version of $v(t)/4$, provided by the feedback loop, with a logarithmic response in amplitude and saturation for high inputs using the nonlinear function

$$F_{LG}(v(t - \tau_c)) = C_1 - C_2 \left(\tanh \left[\log_{10} \left| \frac{C_3}{4} v(t - \tau_c) \right| \right] + 1 \right), \quad (6)$$

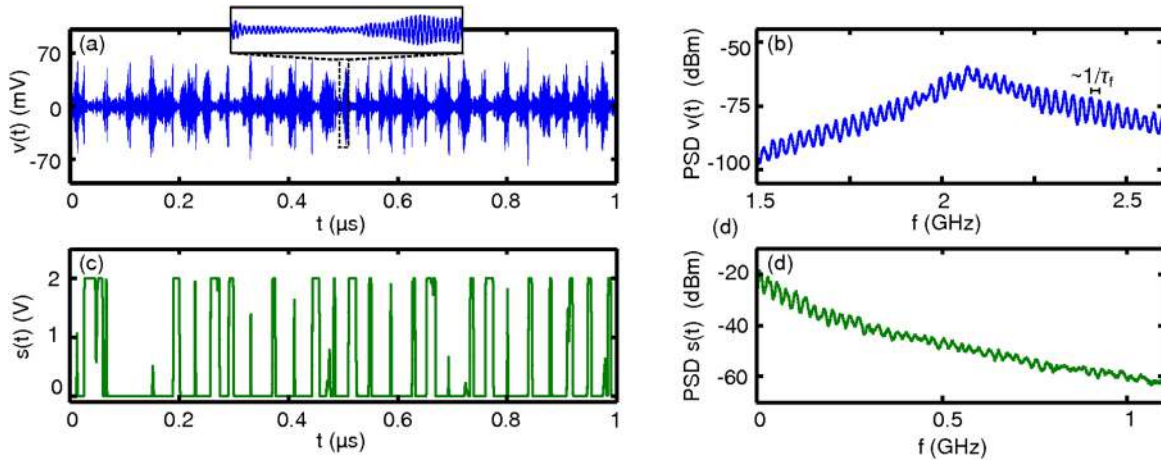


FIG. 6. Dynamics from the model. (a) Time series of $v(t)$ with a zoom shown for $t = 0.5 - 0.525 \mu\text{s}$ and (b) PSD of $v(t)$. (c) Time series and (d) PSD of $s(t)$. The integration was performed using initial conditions for $t \leq 0$: $s(t) = 0$, $\xi(t) = 1.5$, and $v(t) = \text{white Gaussian noise with a variance } \sigma = 10^{-9} \text{ V}^2$ and sampled at a fixed time-step $\delta t = 5 \text{ ps}$. No additional noise is introduced at later times into the model. The PSDs are plotted assuming 50Ω impedance with a frequency resolution of 80 kHz and averaged over a window of 10 MHz.

with parameters $C_1 = 1.68 \pm 0.01 \text{ V}$, $C_2 = 3.24 \pm 0.09 \text{ V}$, $C_3 = 24.1 \pm 5.6 \text{ V}^{-1}$, and $\tau_c = 39.9 \pm 1 \text{ ns}$. To finish, the output of this function drives a first order low-pass filter

$$\dot{\xi}(t) = -2\pi f_L \xi(t) + 2\pi f_L F_{LG}(v(t - \tau_c)), \quad (7)$$

where $f_L = 0.02 \pm 0.01 \text{ GHz}$ is the low-pass cut-off frequency. The output $\xi(t)$ is smoothed to recover the envelope of $v(t)$. This completes the model of the LGA.

The LGA output $\xi(t)$ then drives the TTL gate, which we approximate as a continuous nonlinear switching function

$$s(t) = F_s(\xi(t)) = \frac{A}{2} (1 + \tanh[m(I - \xi(t))]), \quad (8)$$

where $A = 2.00 \pm 0.04 \text{ V}$, $I = 0.96 \pm 0.01 \text{ V}$, and $m = 51.00 \pm 0.01 \text{ V}^{-1}$. Equation (8) switches with rise and fall times proportional to m and asymptotically approaches $s(t) = 0$ and $s(t) = A$ for inputs $\xi(t)$ above and below the threshold I , respectively (Recall that the value of $s(t)/A$ is also used in Eqs. (4) and (5)).

C. Simulation results

1. Numerical time series

We integrate Eqs. (3)–(8) using a third-order Adams-Bashforth algorithm. The simulated time series and frequency spectra for $v(t)$ and $s(t)$ are plotted in Fig. 6, which mirrors the format of Fig. 4 for easy comparison with the experimental results.

As shown in Figs. 6(a) and 6(b), the central frequency of $v(t)$ is $\sim f_c$, and the complex amplitude modulations of $v(t)$ are bounded between $\pm 80 \text{ mV}$. Similar to the experiment, the PSD of $v(t)$ shows a multi-mode power spectrum. The dynamics of the model is thus a good approximation to the experiment, producing a qualitatively similar UHF, multi-mode $v(t)$.

Recall that the model does not include any saturation, and thus, in order for such dynamics in $v(t)$ to remain bounded, the control loop must produce $s(t)$ to switch the gain. In Fig. 6(c), $s(t)$ is an asynchronous, digital-like waveform with some instan-

ces of short-pulse rejection due to the continuous nature of Eq. (8). Different from the experiment, the PSD of $s(t)$ shows small oscillations in its amplitude. We hypothesize that these oscillations are due to high-frequencies in $v(t)$ that bleed through the model's filter for the LGA. A higher order low-pass filter in the model can suppress these oscillations but requires an increase in the model's complexity.

2. Projection of simulated attractor

Next, we examine the attractor and PWL nature of the physical model by constructing time-delayed embeddings using v_n , the local maxima of $v(t)$. Figure 7(a) shows that a 2D embedding forms a projection of the system's attractor with two linear clusters that show qualitative agreement with the experiment, although the clusters produced by the model are less scattered. These quantitative differences occur because our model approximates the system as an ideal two-pole band-pass filter with symmetric rise and fall times in the switching state. Nevertheless, the dynamics of the state (v_{n-T_f}, v_n) in the model transitions between these two clusters as s_n switches states, as shown in Fig. 7(b).

We also examine the simulated probability density $P(g_n)$, given in Fig. 7(c). It shows two primary peaks located at $\sim g_{L,H}$ in qualitative agreement with the experimental results. However, these two peaks have smaller widths due to the assumptions of the model.

Taken together, the time series analyses and attractor projections demonstrate that our PWL model reproduces the fundamental features of the experimental system. Next, by constructing a discrete map that captures all of the complexity of the continuous system, we characterize the stability of the experiment and model to approximate their maximal Lyapunov exponents.

VI. DISCRETE MAP REPRESENTATION AND MAXIMAL LYAPUNOV EXPONENTS

In this section, we construct a time-delayed discrete map to describe the dynamics of the slow-varying amplitude of

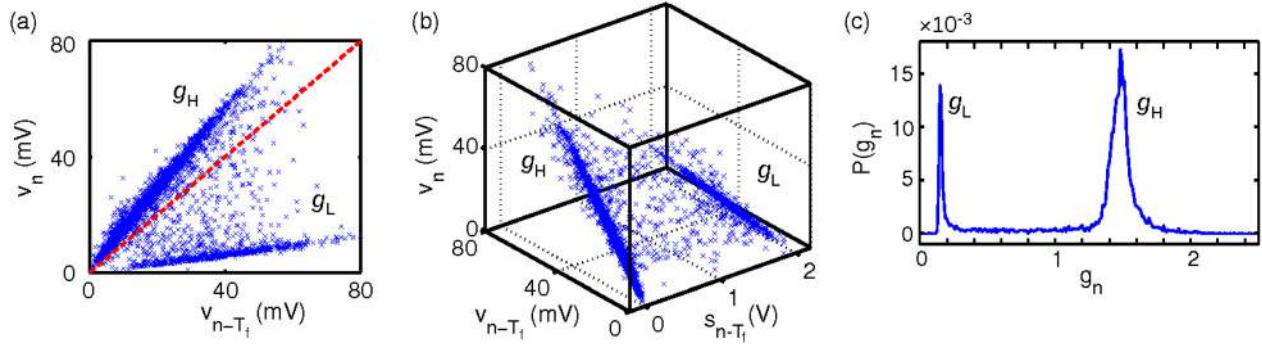


FIG. 7. Simulated attractor projections. (a) 2D and (b) 3D projections of the simulated attractor constructed using s_{n-T_f} , v_{n-T_f} , and v_n , where $T_f = 87$. A dashed line in (a) indicates a reference line with unity slope. (c) The simulated probability density $P(g_n)$ for g_n .

the UHF waveform of our system. As observed in Secs. I–V, the system’s dynamical complexity is not present in the fast oscillations of $v(t)$ but rather in the slow modulation of the local maxima v_n . Thus, we consider the map for the dynamics of v_n , which is a discrete representation of the system. Using this map, we derive a simple expression for the continuous system’s maximal Lyapunov exponent.

To construct the map, we note that the evolution of v_n is determined by v_{n-T_f} , a local maxima in the past with T_f being the average number of local maxima in $[t, t + \tau_f]$, the gain of the system g_n , and the bandwidth $\Delta_{L,H}$ of the VGA. Since the local maxima follow the envelope of oscillations in $v(t)$, the dynamics in v_n is described by a discrete low-pass filter³⁰

$$v_{n+1} = v_n + \alpha_v(g_n v_{n-T_f+1} - v_n), \quad (9)$$

where $\alpha_v = \gamma/(\gamma + \bar{\Delta}^{-1}) = 0.75$, $\bar{\Delta} = (\Delta_L + \Delta_H)/2$, $g_n = (s_n/A)/(g_L - g_H) + g_H$, and $\gamma = 0.5\text{ns}$ is the approximate spacing between local maxima ($\sim 1/f_c$). The switching state of the discrete map is given by $s_n = F_s(\xi_n)$ where ξ_n is the output of a discrete LGA. Equation (9) represents the feedback loop and is approximately PWL.

For the control loop, we construct a discrete version of Eq. (9), which is also a discrete low-pass filter

$$\xi_{n+1} = \xi_n + \alpha_\xi(F_{LG}(v_{n-T_c+1}) - \xi_n), \quad (10)$$

where $\alpha_\xi = \gamma/(\gamma + \omega_{LP}^{-1}) = 0.06$ and $T_c = T_f - 2$ (chosen based on experimental measurements). Thus, Eqs. (9) and (10) with functions $F_s(x)$ and $F_{LG}(x)$ represent our time-delayed discrete map for the local maxima v_n . We plot in Fig. 8 the evolution of the map, its 2D and 3D phase space projections, and the probability density $P(g_n)$ of the discrete feedback-loop gain. These results show similar characteristics to those of the experimental system and its continuous model.

We construct the map’s finite-dimensional Jacobian $(T_f + 1 \times T_f + 1)$ to analyze the stability of a small perturbation $(\delta v_n, \delta \xi_n)$ along the map’s trajectory (v_n, ξ_n) . Its general expression is

$$J = \begin{pmatrix} \frac{\partial v_{n+1}}{\partial v_n} & \frac{\partial v_{n+1}}{\partial v_{n-1}} & \cdots & \frac{\partial v_{n+1}}{\partial v_{n-T_f+1}} & \frac{\partial v_{n+1}}{\partial \xi_n} \\ \frac{\partial v_n}{\partial v_n} & \frac{\partial v_n}{\partial v_{n-1}} & \cdots & \frac{\partial v_n}{\partial v_{n-T_f+1}} & \frac{\partial v_n}{\partial \xi_n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{\partial v_{n-T_f+1}}{\partial v_n} & \frac{\partial v_{n-T_f+1}}{\partial v_{n-1}} & \cdots & \frac{\partial v_{n-T_f+1}}{\partial v_{n-T_f+1}} & \frac{\partial v_{n-T_f+1}}{\partial \xi_n} \\ \frac{\partial \xi_{n+1}}{\partial v_n} & \frac{\partial \xi_{n+1}}{\partial v_{n-1}} & \cdots & \frac{\partial \xi_{n+1}}{\partial v_{n-T_f+1}} & \frac{\partial \xi_{n+1}}{\partial \xi_n} \end{pmatrix} \bigg|_{v_n, \xi_n}. \quad (11)$$

The stability of the map’s trajectories is determined by the magnitude of the largest eigenvalue $|\lambda_n|_{\max}$ of J averaged over the entire attractor. Because of the piecewise-linear nature of Eq. (9), the majority of terms in J are constant or zero, and thus the Jacobian reduces to a simple form:

$$J = \begin{pmatrix} A_1 & 0 & \cdots & \cdots & 0 & A_2 & A_3 \\ 1 & 0 & \cdots & \cdots & 0 & 0 & 0 \\ 0 & 1 & \ddots & & & \vdots & \vdots \\ \vdots & \ddots & \ddots & \ddots & & \vdots & \vdots \\ \vdots & & \ddots & 1 & 0 & 0 & 0 \\ 0 & \cdots & \cdots & 0 & 1 & 0 & 0 \\ 0 & \cdots & 0 & A_4 & 0 & 0 & A_5 \end{pmatrix} \bigg|_{v_n, \xi_n}, \quad (12)$$

where $A_1 = 1 - \alpha_v$, $A_2 = \alpha_v g_n$, A_3 , A_4 , and A_5 are complex functions of the map properties. J can be simplified further by approximating $A_3 \sim 0$ because it is only nonzero when s_n switches ($\sim 5\%$ of points). With $A_3 = 0$, the magnitudes of the eigenvalues λ of J decrease along the trajectories of the map when compared to the true Jacobian’s eigenvalues. Thus, in the context of calculating a maximal Lyapunov exponent, which involves only the eigenvalue of largest magnitude, setting $A_3 = 0$ gives a conservative estimate.

Under this approximation, the eigenvalues λ satisfy the roots of characteristic polynomial

$$(\lambda - A_5)(A_1 \lambda^{T_f-1} - \lambda^{T_f} + A_2) = 0. \quad (13)$$

Based on the factorization of Eq. (13), one of the eigenvalues is always $\lambda = A_5$. Along the map’s attractor, A_5 remains at

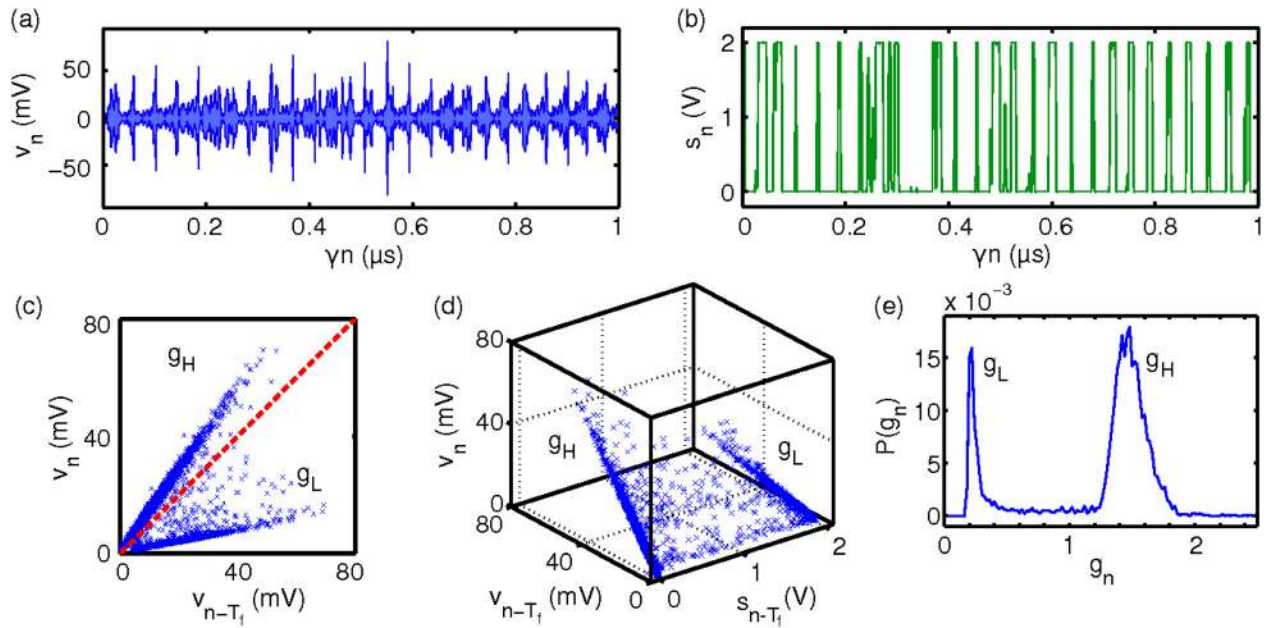


FIG. 8. Piecewise-linear discrete map. Evolution of (a) v_n (solid line) and (b) s_n versus γn ($\gamma \sim 0.5$ ns) with initial conditions: for $n < 0$, v_n = white Gaussian noise with a variance $\sigma = 10^{-9}$ V² and $\xi_n = 1.5$ V. In (a), we also plot $-v_n$ and fill the space in between the two curves for an easier comparison with the amplitude modulations in $v(t)$. (c) 2D projection of the dynamics using v_n and v_{n-T_f} showing two clusters of points with approximate slopes $g_{L,H}$. (d) 3D projection using v_n , v_{n-T_f} , and s_{n-T_f} , demonstrating that s_n partitions the two clusters. (e) Probability density $P(g_n)$ of the gain states in the map's iterated trajectories with peaks labeled $g_{L,H}$.

~ 0.94 V for all n . We determine the other eigenvalues by solving numerically

$$(1 - \alpha_v)\lambda^{T_f-1} - \lambda^{T_f} + \alpha_v g_n = 0, \quad (14)$$

and we find that $|\lambda_n|_{\max}$ switches approximately between two values: $\lambda_L = 0.982$ (locally stable) and $\lambda_H = 1.005$ (locally unstable) for $g_n = g_H$ and $g_n = g_L$, respectively. Since $\lambda_{L,H} > A_5$, these values govern the local rate of maximum expansion.

Interestingly, if we allow $\alpha_v \rightarrow 1$, which removes the dependence of v_{n+1} on v_n (the bandwidth becomes large and Eq. (9) reduces to Eq. (2)), then Eq. (14) becomes $\lambda^{T_f} = g_n$. With this second approximation, which also yields a conservative estimate for $|\lambda_n|_{\max}$, we determine an analytical expression $|\lambda_n|_{\max} = (g_n)^{1/T_f} \sim \lambda_{L,H}$ for $g_n = g_{L,H}$. Thus, as the gain switches, the system switches between stable and unstable trajectories.

Therefore, perturbations in the map's trajectories will grow (decay) if the net-gain is greater (less) than unity. This growth and decay in v_n is a one-dimensional projection of the stretching and folding of the attractor in the full $(T_f + 1)$ -dimensional phase space. Surprisingly, it appears that such a low-dimensional projection is always aligned along the direction of maximum expansion.

Furthermore, based on the similarities when comparing the map's dynamics to the local maxima and switching state in the experiment and model, we conjecture that our circuit's gain switching is also a one-dimensional projection of its attractor's stretching and folding (this assumes that the Jacobian derived from the map is a good approximation for the local flow of the dynamics of the local maxima extracted from the continuous system). Under this assumption, we

now estimate the maximal Lyapunov exponent of the experiment, model, and map.

Given $|\lambda_n|_{\max}$, which are local measurements, the maximal Lyapunov exponent $h_{\max}$ for the trajectories of the system is given by³¹

$$h_{\max} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \ln(|\lambda_n|_{\max}), \quad (15)$$

where $h_{\max} > 0$ is an indication of chaos. Substituting $|\lambda_n|_{\max} = g_n^{1/T_f}$ into Eq. (15) yields

$$h_{\max} = \lim_{N \rightarrow \infty} \frac{1}{T_f N} \sum_{n=1}^N \ln(|g_n|). \quad (16)$$

Assuming ergodicity of the dynamics, Eq. (16) is equal to an integral over the distribution of the system's gain

$$h_{\max} = \frac{1}{T_f} \int \ln(|g_n|) P(g_n) dg_n, \quad (17)$$

where $P(g_n)$ is the probability density of the gain states given by Figs. 5(c), 7(c), and 8(e) for the experiment, model, and map, respectively. The resulting $h_{\max}$ are reported in Table I, showing quantitatively similar, positive values for the experiment, model, and map (using surrogate data of equal length, we have verified that each $h_{\max}$ is significantly positive).

This analysis highlights that our feedback system is fundamentally different from the system in Ref. 13, which only requires a one-dimensional map to solve for its maximal Lyapunov exponent. We acknowledge that a rigorous calculation of the maximal Lyapunov exponent for our continuous system requires a more extensive study to further strengthen

TABLE I. Maximal Lyapunov exponents.

	Experiment	Model	Map	Surrogate data
$h_{\max}$	8.1×10^{-5}	5.3×10^{-5}	7.0×10^{-5}	6.4×10^{-9}

the validity of our approximations. However, we believe these calculations give a deep physical insight into the underlying mechanisms that lead to the chaotic behavior observed in our system.

VII. CONCLUSIONS AND FUTURE DIRECTIONS

In conclusion, we demonstrate a UHF electronic device displaying PWL chaos using feedback and control loops that overcomes many of the hindrances of slower LRC oscillators. We exploit a simple VGA with time delayed feedback to generate growing and decaying multi-mode oscillations centered at frequency ~ 2.1 GHz. Using matched time delays for the feedback and control loops, our circuit performs fast control and avoids nonlinear saturation. As for the dynamics, this novel circuit generates simultaneously a chaotic signal with a corresponding switching state that partitions the system's attractor in 2D and 3D projections, thus, demonstrating the foundation for UHF PWL-chaos with a real-time, readily available symbolic dynamics.

Alongside our experimental system, we presented a continuous model and a discrete map to demonstrate the key aspects from the observed dynamics. We measure the circuit's time-delays and frequency dependent band-pass characteristics and simulate the dynamics using delay differential equations to describe the continuous system in detail and by iterating the map to characterize the system's complexity. These simulations use a logarithmic function, low-pass filters, and continuous switching functions to approximate the piecewise-constant parameters. Our model and map demonstrate a fundamental understanding of the system dynamics and give insight into the non-ideal effects arising in the experiment.

For a balanced comparison, we acknowledge that, unlike the circuit design by Coron *et al.*,^{13,22} our UHF system does not contain an exact analytical solution for the dynamics. In addition, the asynchronous nature of our switching state presents additional challenges for digital communications when compared the exact symbolic dynamics and regularity of their circuit's switching times. Therefore, additional work is still necessary to fully integrate the remaining positive aspects of Ref. 13 into a UHF circuit, and we envision our work as a potential first step for doing so.

In the future, we are interested in applying our UHF design in a radar system. Our system automatically produces an upconverted, broadband analog waveform, and information about this signal can be stored with one-bit digital sampling of $s(t)$. Preliminary analysis shows that the waveforms $v(t)$ and $s(t)$ show potential as radar signal sources when compared to standard techniques.³² We are also currently investigating a matched¹³ (or pseudo-matched³³) filter for chaos that can recover $s(t)$ from $v(t)$ in the presence of large noise. The simplicity of our design combined with these

benefits represent a significant step towards inexpensive and robust chaos-based radars.

Finally, we present only the simplest case for our system where τ_f and τ_c are approximately matched. Preliminary experimental and numerical work shows that additional mismatch between these two delays can give rise to new chaotic dynamics with differently structured PSDs. Thus, the delay mismatch represents an unexplored degree-of-freedom for controlling the dynamics and studying the stability of this system.^{25,34} Potential applications for this additional control include realizing a chaos communications system^{35,36} and creating orthogonal communications channels.³⁷

ACKNOWLEDGMENTS

We gratefully acknowledge the insightful reviewer comments and the financial support of Propagation Research Associates (PRA) Grant No. W31P4Q-11-C-0279.

APPENDIX A: CONTROL LOOP CHARACTERIZATION AND LATENCY TIME

In this section, we characterize the three main components of the control loop: the LGA, the TTL gate, and the voltage control port v_{ctl} of the VGA, in order to estimate model parameters and approximate each component's CLT.

In Fig. 9(a), we plot a typical output waveform from the LGA. Using a signal generator, we drive the LGA with a 2 GHz sinusoidal signal $x(t)$ that is pulsed for 10 ns. When zero power is present in $x(t)$, the LGA outputs a positive voltage $\xi(t) \sim 1.5$ V. For oscillations in $x(t)$, $\xi(t)$ decreases to a value that is proportional to the amplitude of $x(t)$. Hence, the output of the LGA inversely follows the envelope of input oscillations. These time series are measured using two high-impedance probes with identical propagation times after calibration. Thus, the time skew between $x(t)$ and $\xi(t)$ accurately represents the causal behavior of the LGA and shows that the CLT_{LGA} is approximately 8 ns.

Using Eqs. (6) and (7), we drive our model for the LGA using $x(t)$ and fit the output $\xi(t)$ with the model parameters C_i (for $i = 1, 2, 3$) and f_L using a reduced χ^2 algorithm. We note that during the fitting, the CLT of our LGA model is not 8 ns due to the choice of a low-order filter. Thus, extra delay was added into the model LGA's CLT. This extra delay will be discussed at the end of this section. The resulting fit is shown in Fig. 9(b).

We next characterize the TTL gate using the output from the LGA when it is driven by $x(t)$. Using the same high-impedance probes, we measure and plot the TTL output $s(t)$ alongside its input $\xi(t)$ in Fig. 9(c). In the figure, $s(t)$ switches high (low) for $\xi(t)$ below (above) the TTL threshold of ~ 1 V. We mark $\text{CLT}_{\text{TTL}} = 1.1$ ns, the average delayed response of the gate switchings (rise and fall) in response to a threshold crossing.

We use Eq. (8) to fit the TTL gate parameters A , m , and I . The simple model does not produce the correct CLT for the TTL gate, as an additional 1.1 ns time delay is necessary to achieve a good fit (see discussion at the end of this section). The fit is shown in Fig. 9(d). Finally, we

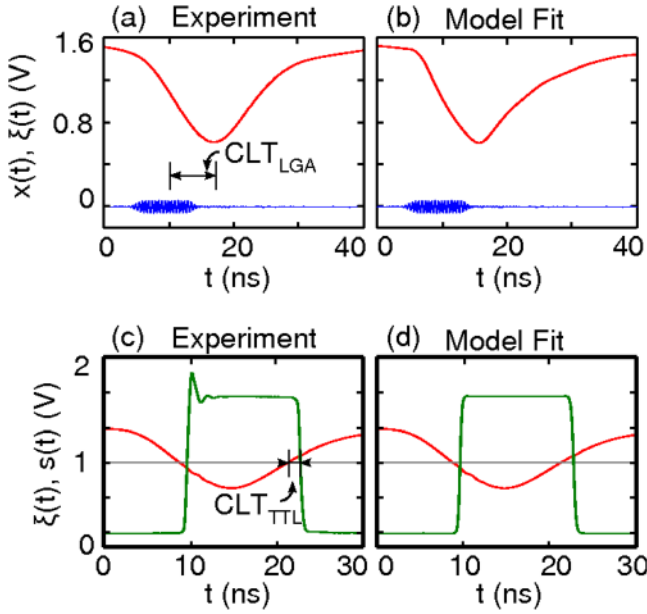


FIG. 9. Control loop fits and latency times. (a) Output $\xi(t)$ (top) of the LGA due to $x(t)$ (bottom). The CLT_{LGA} is shown as a delay time between the two waveforms. (b) Model fit $\xi(t)$ (top) due to input experimental $x(t)$ (bottom). (c) Example output $s(t)$ (square-wave) of the TTL gate due to an input $\xi(t)$ from the LGA. The TTL gate threshold is given by the horizontal bar. A typical CLT_{TTL} is shown for the switching in $s(t)$ from high to low. (d) Model fit $s(t)$ due to input experimental waveform $\xi(t)$.

analyze the processing time of the voltage control port of the VGA in response to a signal from the TTL gate. To do so, we drive the VGA with a 2 GHz low-amplitude sinusoidal signal and observe the output as $s(t)$ switches to measure $\text{CLT}_{v_{\text{ctl}}} \sim 4$ ns. Since the model for the gain of the VGA does not include a processing time, additional delay is necessary in the model.

Therefore, in order to match and model the full control loop delay time τ_c , we use the relation $\tau_c = \text{CLT} + \tau_{\text{coax}}$, where $\text{CLT} = \text{CLT}_{\text{LGA}} + \text{CLT}_{\text{TTL}} + \text{CLT}_{v_{\text{ctl}}}$. We model the coaxial cable delay as a time shift that compensates for any CLT that is not present in our models for the LGA, TTL gate, or voltage control port. As a result, we tune the value of τ_c in the model to match that of the experiment, which is approximately 40 ns.

APPENDIX B: MULTI-MODE TRANSFER FUNCTION AND FITTING

We determine the value of $f_{\text{L,H}}^{(+,-)}$ (used in Eq. (4)) using a fit to the experimental spectral responses shown in Figs. 2(b) and 2(c). To do so, we derive the transfer function for the setup shown in Fig. 2(a) using the differential equation

$$\frac{\dot{v}(t)}{\Delta_{\text{L,H}}} + v(t) + \frac{(\omega_{\text{L,H}}^{(o)})^2 \int v(t') dt'}{\Delta_{\text{L,H}}} = a_{\text{L,H}}(v(t - \tau_f) + v_{\text{in}}), \quad (\text{B1})$$

where $a_{\text{L,H}} = g_a g_{\text{L,H}}$ is the attenuated gain (to keep the system from saturating) and $\Delta_{\text{L,H}}, \omega_{\text{L,H}}^{(o)}$ are the band-pass filter parameters for low (L) and high (H) gain states. We note that

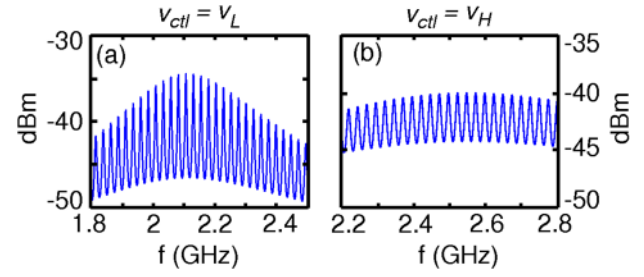


FIG. 10. Using Eq. (B2), we fit the experimental spectra measured in Fig. 2 for (a) $v_{\text{ctl}} = v_{\text{L}}$ and (b) $v_{\text{ctl}} = v_{\text{H}}$.

this equation is similar to that Eq. (3) with an additional input driving term v_{in} and an attenuation g_a in the feedback. We then Fourier transform Eq. (B1) and solve for $H_{\text{L,H}}(f) = \tilde{v}/\tilde{v}_{\text{in}}$, the system's transfer function, where $\tilde{v}$ and $\tilde{v}_{\text{in}}$ are the Fourier transforms of $v(t)$ and $v_{\text{in}}(t)$, respectively. The functional form of $H_{\text{L,H}}(f)$ is

$$H_{\text{L,H}}(f) = \frac{a_{\text{L,H}} \Delta_{\text{L,H}}}{2\pi i f + \Delta_{\text{L,H}} + \frac{(\omega_{\text{L,H}}^{(o)})^2}{2\pi i f} - a_{\text{L,H}} \Delta_{\text{L,H}} e^{2\pi i f \tau_f}}, \quad (\text{B2})$$

where $\Delta_{\text{L,H}} = 2\pi(f_{\text{L,H}}^{(+)} - f_{\text{L,H}}^{(-)})$, and $f_{\text{L,H}}^{(o)} = \sqrt{f_{\text{L,H}}^{(+)} f_{\text{L,H}}^{(-)}}$.

Using the magnitude of $H_{\text{L,H}}(f)$, we fit the experimental data and obtain the fits shown in Figs. 10(a) and 10(b). The fitting parameters are $f_{\text{H}}^{(+)} = 2.41 \pm 0.01$ GHz, $f_{\text{H}}^{(-)} = 1.85 \pm 0.01$ GHz, $f_{\text{L}}^{(+)} = 3.29 \pm 0.1$ GHz, and $f_{\text{L}}^{(-)} = 1.97 \pm 0.04$ GHz. We note that the uncertainties are higher for the frequency parameters $f_{\text{L}}^{(+,-)}$ due to approximation of the resonances in Fig. 2(c) using Eq. (B2). The experimental system is not a true band-pass filter for $v_{\text{ctl}} = v_{\text{H}}$, and our approximation leads to discrepancies between Figs. 2(c) and 10(b), but this approximation is necessary for our simple model. The fitted gain parameters are $a_{\text{H}} = 0.6 \pm 0.01$ and $a_{\text{L}} = 0.2 \pm 0.01$. Also, in order to facilitate fitting, τ_f is a free fitting parameter, where its fitted values are $\tau_f = 41 \pm 1$ ns.

APPENDIX C: TRANSIENT DYNAMICS AND UNDERLYING GRAMMAR

In this section, we compare the experiment, continuous model, and discrete map by examining their transient dynamics as well as the grammar of their potential symbolic dynamics. These comparisons provide more details about the dynamics that were not considered previously.

We plot typical transient dynamics for $v(t)$ and $s(t)$ in Fig. 11. Electrical and additive noise seed the experiment (Fig. 11(a)) and simulations (Figs. 11(b) and 11(c)), respectively, which show multi-mode growth in $v(t)$ and that $s(t)$ begins switching at $0.5 \mu\text{s} < t < 0.75 \mu\text{s}$.

Next, to demonstrate the existence of a potential grammar in the system, we examine four-bit digital words that appear sequentially in the switching state. We sample $s(t)$ at a clock frequency of 1 GHz from the experiment and continuous model and s_n from the map (assuming time steps of γn) and convert the signal into a list of ones and zeros, labeled σ_m : for $\sigma(t_m) > 1$, $\sigma_m = 1$ and for $\sigma(t_m) < 1$, $\sigma_m = 0$, where t_m are the clock sampling times. Next, we construct

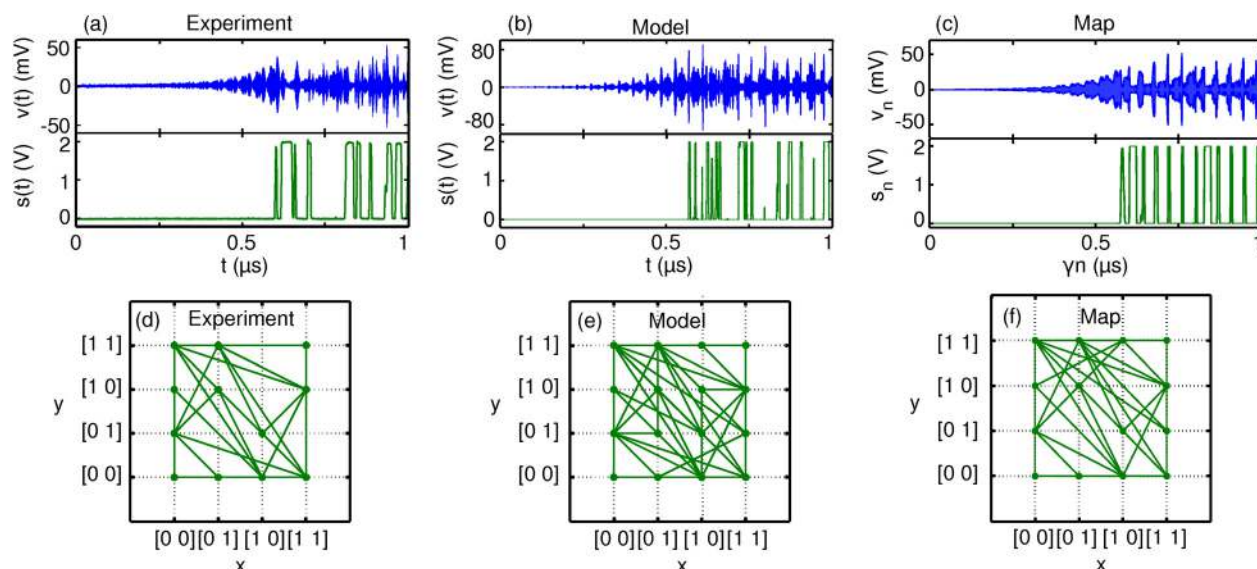


FIG. 11. Transients in $v(t)$ and $s(t)$ for the (a) experiment and (b) continuous model and (c) in v_n (with $-v_n$) and s_n for the discrete map. Four-bit digital word maps for $\sim 10^4$ words in the (d) experiment, (e) model, and (f) map.

the pairings $[0,0]$, $[0,1]$, $[1,0]$, and $[1,1]$ by pairing $[\sigma_m, \sigma_{m+1}]$, $[\sigma_{m+2}, \sigma_{m+3}]$, ... and then follow the evolution of the pairs. We depict the sequences of four-bit words in Fig. 11, where a point is drawn between two pairings on the (x, y) axes if the four-bit word $([\sigma_m, \sigma_{m+1}], [\sigma_{m+2}, \sigma_{m+3}])$ exists in the alphabet. Lines connect sequential four-bit words.

In the analysis of the experiment and map shown in Figs. 11(a) and 11(c), there are missing words and missing links, showing restricted alphabets and grammars. From the continuous model, all words are present but with a restricted grammar. We note that the simulated grammars in the model and map are sensitive to the system's parameters. Thus, we conjecture that mismatches with the experiment lead to the observed differences. Also, we acknowledge that overlapping lines cover missing links and thus a higher-dimensional representation is needed to fully characterize the grammar.

¹Y.-T. Liua, G. Mayer-Kressa, and K. M. Newella, *Acta Psychol.* **103**, 207 (1999).

²Keiji Konishi, Masashi Takeuchi, and Tsuyoshi Shimizu, *Chaos* **21**, 023101 (2011).

³R. Thul and S. Coombes, *Chaos* **20**, 045102 (2010).

⁴J. Vandewalle and L. Vandenbergh, *The Circuits and Filters Handbook*, 2nd ed., edited by W.-K. Chen (CRC, 2003), Chap. 37, p. 1031.

⁵H. Öktem, *Nonlinear Anal.* **63**, 336 (2005).

⁶T. Saito, *Electron. Commun. Jpn., Part. 1* **64**, 9 (1981).

⁷J. N. Schulman, *Phys. Rev. A* **28**, 477 (1983).

⁸T. Saito and K. Mitsubori, *IEEE Circuits Syst.*, **1** **42**, 168 (1995).

⁹D. V. Senthilkumar and M. Lakshmanan, *Phys. Rev. E* **71**, 016211 (2005).

¹⁰F. Han, X. Yu, Y. Feng, and J. Hu, *IEEE Circuits Syst.*, **II** **54**, 1004 (2007).

¹¹A. Buscarino, L. Fortuna, M. Frasca, and G. Sciuto, *IEEE Circuits Syst.*, **I** **58**, 1888 (2011).

¹²K. Srinivasan, D. V. Senthilkumar, K. Murali, M. Lakshmanan, and J. Kurths, *Chaos* **21**, 023119 (2011).

¹³N. J. Corron, J. N. Blakely, and M. T. Stahl, *Chaos* **20**, 023123 (2010).

¹⁴L. O. Chua, *Arch. Elektr. Übertrag.* **46**, 250–257 (1992); available at <http://www-inst.eecs.berkeley.edu/~ee129/sp09/handouts/GenesisChuasCircuit.pdf>.

¹⁵N. J. Corron and J. N. Blakely, *AIP Conf. Proc.* **1339**, 25 (2010).

¹⁶J. N. Blakely and N. J. Corron, *Proc. SPIE* **8021**, 80211H (2011).

¹⁷Y. I. Ismail and E. G. Friedman, *IEEE VLSI* **8**, 195 (2000).

¹⁸A. Tamasevicius, G. Mykolaitis, S. Bumeliene, A. Baziliauskas, R. Krivickas, and E. Lindberg, *Nonlinear Dynam.* **44**, 159–165 (2006).

¹⁹T. Jiang, S. Qiao, Z. Shi, L. Peng, J. Huangfu, W. Z. Cui, W. Ma, and L. Ran, *PIER* **90**, 15 (2009).

²⁰K. M. Myneni, T. A. Barr, B. R. Reed, S. D. Pethel, and N. J. Corron, *App. Phys. Lett.* **78**, 1496 (2001).

²¹Y. C. Koumou, P. Colet, L. Larger, and N. Gastaud, *Phys. Rev. Lett.* **95**, 203903 (2005).

²²N. J. Corron and J. N. Blakely, *Chaos* **22**, 023113 (2012).

²³J. Lü, T. Zhou, G. Chen, and X. Yang, *Chaos* **12**, 344 (2002).

²⁴X. Liu, K.-L. Teo, H. Zhang, and G. Chen, *Chaos, Solitons Fractals* **30**, 725 (2006).

²⁵T. Tsubone and T. Saito, *IEEE Circuits Sys.*, **I** **45**, 172 (1998).

²⁶J. N. Blakely, L. Illing, and D. J. Gauthier, *Phys. Rev. Lett.* **92**, 193901 (2004).

²⁷J. Doyne Farmer, *Physica D* **4**, 366 (1982).

²⁸R. Zhang, H. L. D. S. Cavalcante, Z. Gao, D. J. Gauthier, J. E. S. Socolar, M. M. Adams, and D. P. Lathrop, *Phys. Rev. E* **80**, 045202(R) (2009).

²⁹V. S. Udaltsov, L. Larger, J.-P. Goedgebuier, M. W. Lee, E. Genin, and W. T. Rhodes, *IEEE Circuits Sys.*, **I** **49**, 1006 (2002).

³⁰P. Horowitz and W. Hill, *The Art of Electronics, Second Edition* (Cambridge University Press, 1989), Chap. 9, p. 665.

³¹H. Kantz and T. Schreiber, *Nonlinear Time Series Analysis* (Cambridge University Press, 1997), Chap. 11, p. 175.

³²G. M. Hall, E. J. Holder, S. D. Cohen, and D. J. Gauthier, *Proc. SPIE* **8361**, 836112 (2012).

³³S. D. Cohen and D. J. Gauthier, *Chaos* **22**, 033148 (2012).

³⁴L. Vu and K. A. Morgansen, *IEEE Trans. Autom. Control* **30**, 2385 (2010).

³⁵S. Hayes, C. Grebogi, and E. Ott, *Phys. Rev. Lett.* **70**, 3031 (1993).

³⁶E. M. Bolt, *Int. J. Bifurcation Chaos* **13**, 269 (2003).

³⁷D. Rontani, A. Locquet, M. Sciamanna, D. S. Citrin, and A. Uchida, *Opt. Lett.* **36**, 2287 (2011).